

Project: Deploying a Node.js Web Application on AWS Free Tier Using Docker & Kubernetes

Project Overview

You are tasked with building a simple **Node.js web application**, containerizing it with Docker, storing the image in **Amazon ECR**, and deploying it to a **Kubernetes cluster running on a single EC2 t2.micro instance** using **Minikube**. The application must be publicly accessible via the EC2 instance's public IP address – all within the **AWS Free Tier** limits. No payment is required at any stage.

The application must display basic dynamic information (timestamp, container ID and visitor counter) and be publicly accessible via a web browser.

Primary Goal: Understand the end-to-end workflow of containerization, container registry, and Kubernetes deployment on a single-node cluster, entirely within free tier constraints.

Course: Cloud Computing (BS CS)

Tools: AWS Free Tier, Docker, Kubernetes (EKS or Minikube on EC2), GitHub

Submission Type: Individual

Due Date: June 1, 2026

Deliverables (What to Submit)

#	Deliverable	Format	Submission Method
1	Public Application URL (EC2 IP + NodePort)	Live link	Paste in report
2	Source Code Repository (Node.js app + Dockerfile + Kubernetes YAML)	GitHub (Public)	Link in report
3	Project Report (Max 8 pages)	PDF	Moellim Upload
4	Demonstration Video (5 minutes max)(details given in the end)	YouTube (Unlisted) or Google Drive	Link in report
5	Screenshots (Running app, ECR image, EC2 instance, Minikube cluster, deployed pods)	Embedded in report	Paste in report

Project Report Structure (8 Pages Max)

Your report must contain the following sections:

1. **Title Page** – Project title, your name, roll number, date
2. **Introduction** – What problem does this solve? Why containers + Kubernetes?
3. **Architecture Diagram** – Updated diagram showing EC2 + Minikube + ECR

4. **Step-by-Step Implementation** – Describe each step you followed (use the guideline below)
5. **Challenges Faced** – Minimum 3 issues and how you resolved them
6. **Cost Analysis** – Table showing why you paid \$0 (EC2 free tier, ECR free storage, Minikube is free)
7. **Screenshots** – Minimum 6 screenshots with captions
8. **Conclusion & Learnings** – What did you learn about cloud computing, containers, and Kubernetes?

Step-by-Step Implementation Guideline

Prerequisites (Install on Your Laptop)

Before starting, install these free tools:

- **Node.js** (version 18 or higher) – nodejs.org
- **Docker Desktop** – docker.com/products/docker-desktop
- **AWS CLI** – aws.amazon.com/cli
- **Git** – git-scm.com
- **AWS Account** (Free Tier) – aws.amazon.com/free

Phase 1: Create a Simple Node.js Web Application

Phase 2: Containerize the Application with Docker

Phase 3: Push Image to Amazon ECR

Phase 4: Launch EC2 Instance (Free Tier)

Phase 5: Connect to EC2 and Install Docker & Minikube

Phase 6: Authenticate EC2 to ECR and Pull Image

Phase 7: Deploy to Kubernetes (Minikube)

Phase 8: Testing & Verification

Phase 9: Clean Up Resources

Demonstration Video Requirements (5 minutes max)

Your video must show:

Timestamp Content

0:00 – 0:30 Introduction – your name, project title, explanation of free tier constraints

0:30 – 1:30 Show code on GitHub (Node.js app + Dockerfile + Kubernetes YAML)

1:30 – 2:00 Show Docker image in ECR (AWS Console)

2:00 – 2:30 Show EC2 instance running (AWS Console) and SSH connection

2:30 – 3:30 **LIVE DEMO** – Run `kubectl get pods` and `kubectl get services`, open browser, show running app via EC2 IP and NodePort

3:30 – 4:00 Refresh page, show timestamp changes, show `/health` endpoint

4:00 – 4:30 Show scaling demo (`kubectl scale --replicas=2`)

4:30 – 5:00 Conclusion – what you learned, and demonstrate termination of resources (cost safety)

-Good Luck-